Diandian Peng

Website: https://diandianpeng.github.io | Email: pengdiand@gmail.com | Phone: +1(980)517-0807

EDUCATION

Ph.D. in Geology, University of Illinois at Urbana-Champaign (UIUC) May 2022

Thesis: *Quantifying slab evolution and mantle flow using global subduction models with data assimilation.* Committee: Lijun Liu, Jay Bass, Patricia Gregg, William Guenthner

M.S. in Geophysics, University of Science and Technology of China (USTC) June 2017

Thesis: *Geodynamic modeling of the evolution around the southeastern Tibetan Plateau.*

B.S., School of the Gifted Young, USTC June 2014

Thesis: *A preliminary dynamical simulation for the evolution of the Tibetan Plateau.*

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar, Scripps Institution of Oceanography, UCSD 2022 – 2025

- Studying Tonga subduction zone with mantle convection models
- Modeling the formation of Large Low-Shear Velocity Provinces and mantle plumes
- Investigating plume-slab interaction processes and their geological implications

Graduate Research Assistant, UIUC 2017 – 2022

- Developed global and regional mantle convection models with data-assimilation
- Investigated the formation of special slabs and their influences on the surface
- Studied the seismic anisotropy and slab material migration following mantle flow

Graduate Research Assistant, USTC 2014 – 2017

- Studied the uplift of Tibetan Plateau and its steep boundaries
- Analyzed the crustal channel flow with analytical and numerical methods

Undergraduate Research Assistant, USTC 2013 – 2014

- Explored the evolution of Tibetan Plateau using numerical models
- Studied the thermal structure of continental lithosphere with heat flow data

RESEARCH INTERESTS

- Developing and applying innovative numerical modeling techniques.
- Investigating the interactions between mantle dynamics and tectonic plates.
- Exploring the diverse range of subduction processes and their mechanisms.
- Investigating the formation, evolution, and dynamics of LLSVPs and mantle plumes.
- Studying slab-plume interactions and the associated lithospheric deformation and volcanisms.

PUBLICATIONS

Peer reviewed journal articles

[16] Zhu, T., **Peng, D.** and Liu, L. (2025). High-speed anomaly zone beneath the Changbaishan volcanic province in NE China: Insights from simulation of the Cenozoic Pacific plate subduction using geodynamic models with data assimilation. *Journal of Asian Earth Sciences.* [link]

[15] Zhu, T., **Peng, D.** and Liu, L. (2025). Factors affecting the Pacific plate subduction towards and under the Changbaishan volcanic province since the Cenozoic: Insights from geodynamic modeling based on data assimilation. *Tectonophysics*. [link]

[14] Zhu, T., **Peng, D.** and Liu, L. (2024). Effects of 410- and 660-km seismic discontinuities on the Pacific plate subduction and slab geometry beneath the Changbaishan volcanic province. *Acta Seismologica Sinica* (in Chinese). [link]

[13] **Peng, D.** and Stegman, D.R. (2024). Modeling subduction with extremely fast trench retreat. *Journal of Geophysical Research: Solid Earth.* [pdf]

[12] **Peng, D.** and Stegman, D.R. (2024). Geodynamic evolution of the Lau basin. *Geophysical Research Letters*. [pdf]

[11] Li, Y.[*], Liu, L., Li, S., **Peng, D.**, Cao, Z. and Li, X. (2024). Cenozoic India-Asia collision driven by mantle dragging the cratonic root. *Nature Communications*. [pdf]

[10] Xue, T., **Peng, D.**, Liu, K.H., Obrist-Farner, J., Locmelis, M., Gao, S.S. and Liu, L. (2023). Ongoing fragmentation of the subducting Cocos slab, Central America. *Geology*. [pdf]

[9] Li, Y.[*], Liu, L., **Peng, D.,** Dong, H. and Li, S. (2023). Evaluating tomotectonic plate reconstructions using geodynamic models with data assimilation, the case for North America. *Earth-Science Reviews*. [pdf]

[8] Wang, Y., Cao, Z., Peng, L., Liu, L., Chen, L., Lundstrom, C., **Peng, D.** and Yang, X. (2023). Secular craton evolution due to cyclic deformation of underlying dense mantle lithosphere. *Nature Geoscience*. [pdf]

[7] **Peng, D.** and Liu, L. (2023). Importance of global spherical geometry to model slab dynamics and evolution in models with data assimilation. *Earth-Science Reviews*. [pdf]

[6] Liu, Y.[*], Liu, L., Li, Y.[*], **Peng, D.,** Wu, Z., Cao, Z., Li, S. and Du, Q. (2022). Global back-arc extension due to trench-parallel mid-ocean ridge subduction. *Earth and Planetary Science Letters*. [pdf]

[5] **Peng, D.** and Liu, L. (2022). Quantifying slab sinking rates using global geodynamic models with data-assimilation. *Earth-Science Reviews*. [pdf]

[4] **Peng, D.,** Liu, L. and Wang, Y. (2021). A newly discovered Late-Cretaceous East Asian flat slab explains its unique lithospheric structure and tectonics. *Journal of Geophysical Research: Solid Earth*. [pdf]

[3] **Peng, D.,** Liu, L., Hu, J., Li, S. and Liu, Y.[*] (2021). Formation of East Asian stagnant slabs due to a pressure-driven Cenozoic mantle wind following Mesozoic subduction. *Geophysical Research Letters*. [pdf]

[2] Liu, L., **Peng, D.,** Liu, L., Chen, L., Li, S., Wang, Y., Cao, Z. and Feng, M. (2021). East Asian lithospheric evolution dictated by multistage Mesozoic flat-slab subduction. *Earth-Science Reviews*. [pdf]

[1] **Peng, D.** and Leng, W. (2017). Analytical and numerical simulations of uplift processes at the Tibet-Sichuan boundary. *Earthquake Science*. [pdf]

Preprint articles

[1] Suo, Y., Dong, H., Liu, L., **Peng, D.,** Li, Y.[*], Liu, J., Dai, L., Cao, X. and Li, S. (2022). Landward mantle flow associated with the Pacific subduction system opened the South China Sea. [link]

Books

[1] Li, S., Zhu, J., Cao, X. Liu, L., Zhou, J., and **Peng, D.** (2023). *Atlas of global mantle microplates with tomographic images*. Science Press (Beijing). ISBN: 9787030752086. (Selected as one of the "Outstanding Marine Science and Technology Books for 2024" by the Chinese Society for Oceanography.)

Articles in-prep

[2] **Peng, D.**, Stegman, D.R., Day, J.M.D., Parnell-Turner, R., Liu, L. and Wang, Y. Intraplate Indian Ocean magmatism and the Toba super-volcano produced by the Capricorn plume.

[1] **Peng, D.**, Liu, L., Cao, Z., Li, Y. *, Stegman, D.R. and Wang, Y. Cenozoic uplift of western US due to removal of dense lithospheric roots.

* Students mentored

GRANTS

TACC. Simulating slab-plume interaction with realistic mantle convection models. PI: Dave Stegman

Awarded **224640 SUs** (12.6 million core-hours) on TACC Frontera. 2025

Awarded **230520 SUs** (12.9 million core-hours) on TACC Frontera. 2024

Awarded **225000 SUs** (12.6 million core-hours) on TACC Frontera. 2023

***DP** contributed to the scientific justification, usage plan, writing the proposal, and executed the work on Frontera.*

UCSD. IGPP Green Scholar Award. **$138,000**. PI: Diandian Peng 2022 – 2024

NSF EAR 2244660. Investigating formation of stagnant slabs and implications for subduction dynamics. **$333,141**. PI: Lijun Liu and Craig Lundstrom 2023

***DP** contributed to the scientific justification and figures of the proposal.*

NSF OAC 2031704. NSF Frontera Allocation Travel Grant. **$10,000**. PI: Lijun Liu 2020

***DP** contributed to the scientific justification and figures of the proposal.*

HONORS AND AWARDS

Cecil H. and Ida M. Green Postdoc Fellowship (UCSD) 2022

Bluestem Fellowship (UIUC) 2017

Outstanding Student Paper Award (CGU) 2015

Multiple Undergraduate/Graduate Scholarships (USTC) 2010 – 2016

SELECTED CONFERENCE ABSTRACTS

Liu, L., **Peng, D.,** and Li, Y. * (2023). What drives uplift of west-central North America since the Late Cretaceous? In *AGU Fall Meeting Abstracts.*

Li, Y. *, Liu, L. and **Peng, D.** (2022). What drives the post-collisional northward Indian motion? In *AGU Fall Meeting Abstracts*

Peng, D., Liu, L., Chen, L. and Li, S. (2021). East Asian lithospheric evolution in response to west Pacific subduction since 100 Ma, In *AGU Fall Meeting Abstracts.*

Peng, D. and Liu, L. (2021). Quantifying horizontal vs vertical motions of subducted slabs using global geodynamic models with data-assimilation, In *AGU Fall Meeting Abstracts.*

Liu, L., **Peng, D.,** Liu, L., Li, Y. * and Chen, L. (2021). A new geodynamic framework for the evolution from intraplate orogeny to continental extension, In *AGU Fall Meeting Abstracts.*

Li, Y. *, **Peng, D.,** Cao, Z. and Liu, L. (2021). Continental-scale converging mantle flow beneath Eurasia and America, from model to observation, In *AGU Fall Meeting Abstracts.*

Liu, L., Zhou, Q. and **Peng, D.** (2021). Cenozoic topographic evolution of western-central US, In *GSA Annual Meeting.*

Peng, D., Liu, L., Chen, L. and Li, S. (2020). Late-Cretaceous Izanagi flat subduction below East Asia and tectonic responses. In *AGU Fall Meeting Abstracts.*

Liu, L., **Peng, D.,** Wang, Y. and Hu, J. (2020). Differences and similarities of lithosphere deformation during flat-slab subduction and plume underplating. In *AGU Fall Meeting Abstracts.*

Peng, D., Liu, L., Liu, Y. *, Hu, J. and Li, S. (2019). Formation of East Asian stagnant slabs due to a westward Cenozoic mantle wind. In *AGU Fall Meeting Abstracts.*

Liu, L., Hu, J., Zhou, Q., Liu, Y. and **Peng, D.** (2019). Formation of slab tears controls Earth's major silicic volcanisms. In *AGU Fall Meeting Abstracts.*

Peng, D., Liu, L. and Hu, J. (2018). Understanding subduction dynamics in the Southwest Pacific. In *AGU Fall Meeting Abstracts.*

Peng, D. and Leng, W. (2016). Analytical and numerical simulation of uplift at the Tibet-Sichuan boundary. In *AGU Fall Meeting Abstracts.*

* Students mentored

TEACHING AND MENTORING

Mentoring experience at UIUC

Yiming Liu (Visiting graduate student at UIUC)

Guided research led to a publication in Earth and Planetary Science Letters.

Yanchong Li (Graduate student at UIUC)

Guided research led to two publications in Earth-Science Reviews and Nature Communications, along with several abstracts presented at AGU Fall Meetings.

Teaching experience at UIUC

Spring 2021: GEOL 104 — Geology of the National Parks

Role: Grade homework and provide feedback

Fall 2018, Fall 2020: GEOL 117 — The Oceans

Role: Grade homework and provide feedback

Teaching experience at USTC

Fall 2016: Introduction to Earth Science

Role: Grade homework, provide feedback and answer questions

FIELDWORKS

R/V Thomas G. Tompson cruise in the South Pacific Ocean — Nov – Dec 2023

- *Successfully installed and deployed Ocean Bottom Seismometers (OBS) near the Lau Basin and Samoa Hotspot to collect seismic data for geophysical research.*
- *Conducted dredging operations near the Samoa Hotspot and Cook Islands to obtain rock samples for geochemical and petrological analysis.*
- *Performed precise cutting, classification, and detailed description of rock samples.*

Field work in Baraboo (WI, US) — 2021

Field work in Sugar Creek, Kentland and Williamsport (IN, US) — 2017, 2019

Field work in Chaohu (Anhui, China) — 2014

BLOG POSTS

Why do some slabs stagnate? *EGU Blogs*. [link]

SERVICES

- Department colloquium coordinator (UIUC, Fall 2018)
- Volunteer for *Engineering Open House* (UIUC)
- Reviewer for *Geoscience Letters, Geophysical Research Letters, Geology, G-Cubed, Scientific Reports, Proceedings of the Royal Society A, Physics of the Earth and Planetary Interiors*

INVITED TALKS

- University of South Florida. Where the mantle wind blows. 2025
- Nanjing University. Applications of mantle convection models across diverse subduction zones and insights into overriding plate deformation. 2024
- Southern University of Scientific and Technology. Studying subduction zone dynamics using numerical models with different dimensions. 2024
- Chinese University of Hong Kong. Studying circum-Pacific subduction zones with geodynamic models. 2023
- Scripps Institution of Oceanography. IGPP Seminar Series. 2023